

CIS 700 Final Project - AI The Asshole: An Interactive Platform to study Machine Ethics

Karmanya Aggarwal, Jeffrey Cho, Elizabeth Dinella, Josh Ludan

17 April 2022

1 Introduction

With the incredible success of transformer language models in conversation [5, 7, 14], concerns have emerged regarding unethical implicit biases. Language models periodically output abusive language as racism, misogyny, Islamophobia, homophobia, and ableism tend to be overrepresented in training data [3]. As such, interest and research in *Machine Morality* has grown. Many groups including Google, Microsoft, IBM, and Samsung are devoting resources to study responsible and ethical AI practices [26]. In 2021, the Allen Institute for AI released Delphi [17], a significant step toward modelling morality. However, researchers in AI ethics [24] argue that ethical judgments are dynamic and cannot be derived from a set of existing judgments.

Modelling a static dataset collected from a historic/snapshot corpus has been criticised in various domains [19]. In his book, *Rebooting AI*, Gary Marcus argues:

A system that reads news stories, for example, can't be trained on what happened last week or last year or even in all recorded history, because new situations pop up all the time.

As an alternative to static offline datasets, *online learning* is a technique to learn from continuous streams of data [16]. Online learning allows the model to be updated as new training data arrives. In domains where continuous data streams are not naturally present, model designers often turn to *crowdsourcing*. Crowdsourcing refers to mass work done by experts or nonexperts [25]. Workers are tasked with explicitly generating data or more abstract tasks to increase trust in AI systems (ex. human in the loop learning). Regardless of the tasks, crowdsourcing is a valuable tool to allow AI systems to continuously learn and keep up with an ever-changing world.

In this project, we study machine morality through an interactive experience. As a large source of interactive discussions on morality, we study the **AmITheAsshole** subreddit [23]. **AmITheAsshole** (AITA) is a popular online community with 3.7 million members. The subreddit is a forum in which users submit short stories about interpersonal conflicts they've experienced. The posters solicit responses to judge if their actions were appropriate, or if they are the "asshole" in the conflict.

As described by the community's moderators:

A catharsis for the frustrated moral philosopher in all of us, and a place to finally find out if you were wrong in an argument that's been bothering you. Tell us about any non-violent conflict you have experienced; give us both sides of the story, and find out if you're right, or you're the asshole.

In this project, we provide a real-time interactive experience between a human user acting as the judge and an AI system acting as the poster. Our game play mimics the experience between poster and commenters on the subreddit. The user "plays" as a judge of the generated conflict and determines if the storyteller is the "asshole" in the conflict.

Our proposed contributions are the following:

1. An artifact of our interactive experience.
2. A platform for crowdsourcing morality labels.

3. An in depth description of our methodology and challenges we experienced along with our solutions.
4. An extensive evaluation of our model’s conversational effectiveness including UNION/Bert-score and crowdsourced human evaluation.

2 Related Work

Dathathri et al. introduced the Plug and Play Language Model (PPLM) for controllable language generation[8]. The model uses simple attribute classifiers to guide text generation by combining to a pretrained language model. Guided text generation can be performed without fine tuning by treating conditional probabilities together. This study showed a novel way to enhance guided text generation using a simple approach, and GPT-2 based evaluation discussed in this paper can be easily adapted to GPT-3 based applications.

Alshomary et al. solved the counter argument generation problem by two step strategy[1]. First they detected and ranked premises by the strength of the argument. Second they generate counter arguments for the weak premises. Pinpointing the weakest argument is a great approach to this problem, which makes counter argument generation easier and more robust.

Emelin et al. presented Moral Stories, which is a goal-oriented short narrative that moral reasoning is grounded in situations[10]. They also evaluated models on classification and generation tasks based on the dataset. They also introduced Chain-of-Experts, decoding strategies that enforce constraint satisfaction to NLG models. This study showed multiple aspect of view of moral problems inherent in text generation. Classification model can check whether large language model can understand moral sense, and generation model can output morally guided texts, which can be used as counter arguments.

Recently, AITA subreddit has been actively studied by various researchers. Haworth et al. trained machine learning models on AITA posts for classifying reasonability[15]. Botzer et al. and Efstathiadis et al. analyzed moral judgments on AITA articles by training a classification model[4, 9]. Many studies are focusing on detecting and classifying moral judgement with AITA dataset, which shows the subreddit is a good resource of training morality. On the other hand, our work reveals how GPT-3 model can understand morality with prompted and fine-tuned examples, and examine if generated comments with opposite stance can effectively convince human judgement.

3 Game Play Description

Our interactive platform can be thought of as a “game” in that a user has a goal, and reaching that goal provides an entertaining experience. Our game will have an infinite series of rounds where each round will correspond to generating a single data point. For each round we will generate a list of “links” containing post titles that the user can select. Upon selection of a post title, a short story based on the selected title will be generated by a language model and presented to the user. The player has the task of classifying the generated story as YTA (you’re the asshole) or NTA (not the asshole). The player will then be scored based on whether they made the “right” decision or not. Occasionally, a player will be “challenged” by another generative model simulating a commentator. This commentator will provide a summary of a view that opposes whatever decision the player chose. The player will then be able to revise their decision after receiving this challenge. The likelihood of this challenge is higher if the player made the wrong decision but if the player made the right decision they still have a chance of being challenged. The “challenge” game play component is included as a quality control mechanism. Judging morality often involves considering multiple viewpoints before reaching a decision. We generate an opposing viewpoint to force the player to consider alternate judgements.

We show example game interface for generated stories and comments in Figure 2 and Figure 3 in Appendix.

4 Dataset

The popular online form, Reddit, is frequently used as training data [12, 11, 13]. Reddit provides a good source of data as it is public / open-access and has a nice subreddit structure [22]. Finding relevant data is easier than say, tweets which are mostly unstructured.

Previous work [20] collected a dataset of 63k samples of the form $(post, label)$ where $label \in \{YTA, NTA\}$. However, the provided dataset does not provide the entire comments in textual form. The comments are only implicitly included in the label. The “challenge” component of our gameplay will likely require modelling of comments. Unfortunately, the existing dataset does not include any metadata necessary to augment with comment data. As such, we set out to collect our own dataset of $(post, comments)$ pairs where comments are the text of the top-k comments ranked by number of upvotes.

4.1 Dataset Collection

To collect a large dataset of posts and comments from the AITA subreddit we used pre-processed dataset [21] that gathered with Pushshift API [2] which provides developer tools for scraping Reddit. Originally the dataset has 97,628 AITA articles. Each sample contains a body, title, and a YTA/NTA label. We filter the comment labels for simplicity. The subreddit’s judgement decisions are non-binary. Responses can include tags such as: ESH = Everyone Sucks here; NAH = No A-holes here; INFO = Not Enough Info. However, our dataset and gameplay only include the binary labels “YTA” or “NTA” for simplicity.

For fine-tuning and evaluating our model, we further filtered and balanced articles for labels. While we including posts to our sample set, articles that have more upvotes are preferentially considered. By this way we could include highest quality posts. As the result, we maintained 9,965 instances which have balanced labels for YTA and NTA. For this sample dataset, we additionally scrapped all corresponding comments for story generation and comment generation.

4.2 Dataset Statistics

In our sample dataset, there are 17.9, 426.8 tokens in average for titles and bodies. For each article, there are 547.2 comments in average. We are showing histogram of number of tokens in title and body, and number of comments in articles in Figure 1. We also report base statistics of our sample dataset in Table 1.

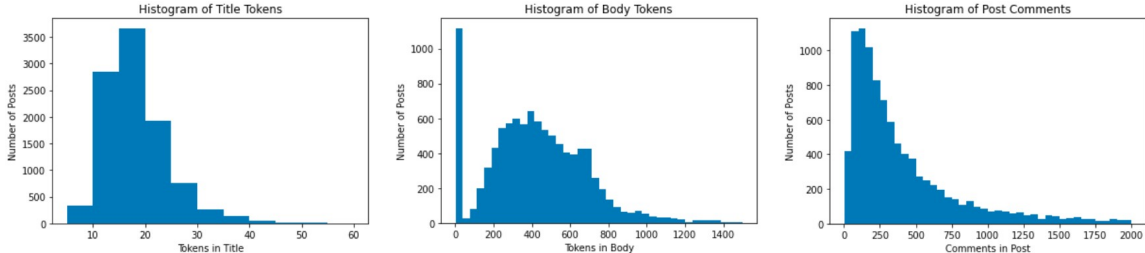


Figure 1: Histogram of number of tokens in title and body, and number of comments in articles

Label		Original (n = 97,628)		Sample (n = 9,965)	
NTA	Not the A-hole	59,114 (60.5%)	72.8%	4454 (44.7%)	50%
	No A-hole Here	12,003 (12.3%)		528 (5.3%)	
YTA	You are the A-hole	20,959 (21.5%)	27.2%	4,019 (40.3%)	50%
	Everyone Sucks Here	5,552 (5.7%)		963 (9.6%)	

Table 1: Base statistics of original and sample AITA dataset.

5 Method

Our method involves multiple independently trained language models.

1. **Story generator model:** This model will generate the short “post” stories. We model:

$$P(x|l, t)$$

where x is the post (and title), l is the label (either an NTA or a YTA based on the commenter’s consensus), and t is a topic. That is, given a prompt which specifies whether the poster is in the wrong, our model is tasked with generating a corresponding story. Our technique is to fine tune a GPT3 curie language model on 1.5k samples. The input prompt is constructed by instantiating t as a set of comments. We also experimented with t as a set of keywords but found it to perform empirically worse.

2. **Story summarize model:** This model will be tasked with generating a “challenging” viewpoint to the player’s initial judgement decision. Here, we aim to model:

$$P(c|x, l)$$

where c is a comment with a judgement of l . That is, the model takes an AITA post and YTA/NTA flag as input, generates a oppositely biased summarization of original posts that counters the input flag. We implement our approach by a finetuning GPT curie model for each l on 2k samples. That is, two seperate models are finetuned: $P(c|x, YTA)$ and $P(c|x, NTA)$

Example prompts are shown in Figure 5.

Prompt	Generation
Title: AITA for joking about my wife’s cheating? Label: YTA	Body: I was at a party with some friends and I got drunk. I started joking about my wife cheating on me. I said things like ”I can’t believe she cheated on me with that guy” and ”I bet she’s cheating on me right now”. I was just joking around but some people at the party thought I was serious and they started comforting me. I tried to tell them I was joking but they didn’t believe me. My wife found out about this and she’s really mad at me. She says I embarrassed her and she doesn’t want to be around me. AITA?
Title: AITA for making fun of my friends weight? Label: YTA Body: My friend is a bit overweight. I am not, but I am not skinny either. We were talking about our diets and she said that she was trying to lose weight. I said that I was trying to gain weight. She said that I didn’t need to because I was already ”fat enough”. I laughed and said ”You should talk!” She didn’t find it funny and she said that I was being rude. I told her that I was just joking and she said that it wasn’t a funny joke. I don’t understand why she was so offended.	Top Replies: 1) NTA, that was one hell of a joke and I congratulate you for it. 2) NTA, it’s a fun joke but wrong audience.. 3) NTA. A+ for execution

Table 2: Example prompts for the Story Generator Model and the Story Summarization Model. The first row is the prompt for the Story Generator Model. The input includes the title and a binary YTA label. The model is tasked with generating the output body in the second column. The second row is the prompt for the Story Summarization Model. The input includes a title, label, and body. The output generation a sequence of comments that challenge the input label.

6 Experiments & Performance Metrics

For the story generation models, we experimented with 2 kinds of generation, first by using comments from posts from the actual subreddit to prompt the model to generate a story. For this

model, the stance of the generated story (YTA/NTA) is controlled by selecting a set of appropriately stanced comments from that post. In this approach, a single curie model is trained on a large subset of posts and the prompt alone controls the stancing of the generated text. The second approach involves training 2 curie models individually such that one model generates YTA stories and the other generates NTA stories. Further this approach involved extracting keywords from stories in the training set which was done via YAKE[6] and using just these keywords as the prompt to generate stories. In order to ensure that the models are comparable, we attempted to keep the fine-tuning budget for both approaches equal, that is, the single model cost twice as much as either of the two individual models to fine-tune. Human evaluation of a set of 40 stories from both approaches indicates that the first approach performed better and was used in the final gameplay.

For the summarization models, we experimented with 2 approaches - a direct approach that was tuned on the highest scoring NTA/YTA posts from our training set, and an ‘argumentative’ approach, that was tuned on the highest scoring oppositely stanced comment from a stanced post. For example, in a story that was marked YTA by the subreddit’s users, we picked the highest scoring NTA post, and vice versa.

6.1 Automated Evaluation

Generation of stories are traditionally difficult to evaluate as we’ve already seen in various papers in class. Though previous work in this field has attempted to use a variety of metrics, it’s currently unclear if a high score on a singular or set of metrics correlates strongly to what humans would consider an enjoyable story. However, automated metrics definitely have a place in that they accelerate the speed of experiments and that the models that achieve the best automated evaluation will be passed onto human evaluation. We plan on use the following automated metrics -

1. Rouge-L
2. Perplexity, using gpt2 as the base model for calculating perplexity
3. BLEU - n-gram overlap of target story and generated story. This is not a great metric simply because the ability to reproduce a story seen in the training set doesn’t necessarily mean that the model will generalize to novel stories.
4. Bigram Degree of Repetition - proportion of stories that repeat at least 1 bi-gram
5. Distinctness - Similar to [18], we measure the ratio of distinct trigrams to the total number of trigrams in the generated story. Though [18] uses 4 grams, we found that even for the baseline models, there were such few repeated 4 grams that the metric wasn’t particularly useful.
6. BertScore [27] calculated between reference stories/comments and the model generated stories/comments

6.2 Human Evaluation

We performed extensive Human Evaluation on the best performing models from the automated metrics. Firstly, annotators were provided a small set of stories and asked to flag stories that were either incoherent, flawed in a way that they are unusable (grammatically incorrect, racist etc). They were also asked to gauge if the stories were stanced correctly, that is- did the model correctly generate a story where the typical subreddit user would vote ”YTA” or ”NTA”. Secondly, annotators were provided with pairs of stories such that a singular story from each pair could have originated from our gold set of reddit stories, or from the comment-based model, or from the keyword-based models. For each pair of stories, annotators were tasked to pick which story they preferred based on the following criteria-

1. Grammatical correctness and completeness
2. Internal consistency/coherence
3. In cases of ties, enjoyment of the story

Treatment	Story A Preferred	Story B Preferred
Reddit(A) vs Reddit(B)	9	11
Model(A) vs Reddit(B)	1	39
Comment(A) vs Keyword(B)	19	14

Table 3: Pairwise Evaluation of the two approaches to story generation

	Comment Based Model		Keyword Based models	
	YTA	NTA	YTA	NTA
Proportion that Matches Tag	0.5 (12)	0.72 (18)	0.29 (14)	0.61 (13)
Proportion Coherent	0.4651 (13)		0.30 (27)	
Proportion Acceptable	0.77 (13)		0.59 (27)	

Table 4: Qualitative evaluation of the Story Generation Approaches. Since the number of annotated stories was quite small, the actual number of qualifying stories is presented to relay effect size

We understand that ‘enjoyment’ of a story is rather subjective and difficult to specify - however, given the nature of this interactive experience, we believe that it’s important to evaluate. Thus, as a control, a small number of pairs consisted only of stories taken from the gold set, in the hope that this set would reveal any obvious biases in the annotators choices.

To evaluate the summarization generation models, 2 annotators were presented a set of 10 stories from the comment-based generation model, along with a YTA stanced summary as well as an NTA stanced summary from each of the 2 approaches, and asked to pick which summary they preferred, based on the same criteria as the summarization model.

7 Evaluation

We experimented with few-shot learning using the GPT3 D model. Specifically, we tried to see if we would be able to generate good stories using the replies as a prompt. A small scale qualitative analysis revealed that the model could generate coherent interpersonal conflicts. However, we found it difficult to control the NTA/YTA labels. In several runs, we found that YTA stories were generated despite an NTA-biased prompt and vice versa. Select examples of generated stories can be found in the attached table. The second and fourth example generation represent examples where the generated story labels don’t align with the prompt label. We present automated evaluation results for the story generation models in Table 5. In the table, Curie-1500 represents a GPT3-Curie Model finetuned on 1500 story/comment pair samples, and Davinci-50 represents a GPT3-Davinci Model finetuned on 50 story/comment pair samples.

Given the efficacy of the Curie models, and their significantly cheaper cost, we used curie as the base model for both the story generation approaches as well as the 2 summarization approaches. Table 4 shows the results of the qualitative human evaluation of the two story generation approaches. Since the total number of stories annotated was quite small, the total number of stories that fall into the categories are also presented to give an idea of effect size. Despite the small number of evaluated stories, the first model, ie the comment-based model outperforms the keyword-based models. In the pairwise evaluation, our annotators were presented with 20 pairs such that both stories were taken from the gold set, 40 pairs such that one story was taken from the gold set and the second was generated by a model, and 40 pairs such that each story was generated from a different model. Stories from various approaches were shuffled to prevent a bias from influencing the results (Eg, a tendency to break ties by selecting Story A) Results from Pairwise evaluation are presented in Table 7

For the summarization models, we evaluate a GPT-3 curie baseline on our automated evaluation metrics and present results in Table 6. For human evaluation, 2 annotators were presented with 10 stories, 5 from each stance, and stanced summary from the NTA comment generation models. Of the 10 stories, our annotators agreed on 6/10 samples, and in 5/6 strongly preferred the summaries generated from the direct approach, thus that was used in the final gameplay.

Metric	Curie-1500 Finetune	Davinci-50 FineTune
Rouge-L - fmeasure	0.19	0.21
Bleu	0.06	0.06
Average BertScore	0.84	0.85
Gold Set Perplexity	10.98	11.37
Predicted Set Perplexity	25.33	2.09
Gold Set Degree of Repetition	1.0	1.0
Predicted Degree of Repetition	1.0	1.0
Gold Set Distinctness	0.97	0.97
Predicted Distinctness	0.87	0.86

Table 5: Story Generation Evaluation Metrics

Metric	YTA Summaries	NTA Summaries
Rouge-L - fmeasure	0.12	0.15
Bleu	0.007	0.008
Average BertScore	0.84	0.84
Gold Set Perplexity	12.53	80.52
Predicted Set Perplexity	15.50	28.73
Gold Set Degree of Repetition	0.60	0.51
Predicted Degree of Repetition	0.78	0.73
Gold Set Distinctness	0.99	0.99
Predicted Distinctness	0.94	.95

Table 6: Baseline GPT3-curie model performance on the comment generation subtask.

8 Discussion

From our experiment, we have seen that GPT-3 is capable to perform required functions in our text game. First of all, it can generate reasonable and interesting AITA titles with real examples as prompt. Most of them are dramatic situations which are possibly happening in real life. Second, after fine-tuning, with the guidance of sample comments, GPT-3 model can usually generate plausible and logical AITA bodies. Third, GPT-3 is able to generate comments to an AITA article as counter opinions, which worked as summarization with guided biased stances. With the combination of three applications above, our game is fully functioning and interesting to play.

However, we also encountered some problems during our analysis. To begin with, story generation model are heavily guided by the prompt comments. If the comments selected have clashing details without any clear way to reconcile them, it leads to story incoherencies as GPT-3 fails to meaningfully transition and connect parts of the story together. Readability of generated stories is one of the fundamental challenge of GPT-3 model, and we expect our game will be more stable and realistic if this issue can be solved.

Anecdotal Commonsense and Logical Errors During the human evaluation, we found a number of occasions in which our model(s) generated stories that did have “commonsense.”

For example, the commonsense fact “Your daughter be more than 5 years younger than you” is violated in the following quote from our story generation model: `My (17f) daughter (12) is a very good girl, she’s a little anxious but she’s doing well in school and overall just a good kid.`

In a different story, the model violates the fact “You cannot buy US \$ bills at a convenience store.” `Yesterday, we were running low on cash and we went to a convenience store to get some. We bought a 20$ US bill.`

In another generated story, the model violates the social assumption that a “dad” with he pronouns is not a woman: `I’ve been with my SO for 4 years and we live together. I am a veterinary technician and I work long hours and he is a stay at home dad to our daughter. My parents are very traditional and they don’t approve of our relationship because we’re both women.`

Moreover, some of the original AITA articles may include sexual and violence contents. After fine-tuned, GPT-3 model sometimes generates unsafe topics or story details that shouldn’t be dealt

with. Stopwords filtering could be one of the option that we can tackle this problem, but it cannot cover all edge cases, and this method often filter some good articles. Inappropriate text generation is one of the key concern of OpenAI, and this issue should be carefully treated for avoiding unnecessary offence to players.

9 Attribution

Karmanya Aggarwal: Implementing comment summarization model, automated story generation evaluation and human annotation tools.

Elizabeth Dinella: Implementing comment summarization model and fine-tuning.

Josh Ludan: Implementing story generation model and story quality assessment, creating game interface, data gathering and pre-processing.

Young-Min(Jeffrey) Cho: Implementing story generation model, creating game interface, doing literature review.

References

- [1] Milad Alshomary, Shahbaz Syed, Arkajit Dhar, Martin Potthast, and Henning Wachsmuth. Argument undermining: Counter-argument generation by attacking weak premises. *arXiv preprint arXiv:2105.11752*, 2021.
- [2] Jason Baumgartner, Savvas Zannettou, Brian Keegan, Megan Squire, and Jeremy Blackburn. The pushshift reddit dataset, 2020.
- [3] Emily M. Bender, Timnit Gebru, Angelina McMillan-Major, and Shmargaret Shmitchell. On the dangers of stochastic parrots: Can language models be too big? . In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency, FAccT ’21*, page 610–623, New York, NY, USA, 2021. Association for Computing Machinery.
- [4] Nicholas Botzer, Shawn Gu, and Tim Weninger. Analysis of moral judgment on reddit. *IEEE Transactions on Computational Social Systems*, 2022.
- [5] Tom Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared D Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, et al. Language models are few-shot learners. *Advances in neural information processing systems*, 33:1877–1901, 2020.
- [6] Ricardo Campos, Vítor Mangaravite, Arian Pasquali, Alípio Jorge, Célia Nunes, and Adam Jatowt. Yake! keyword extraction from single documents using multiple local features. *Information Sciences*, 509:257–289, 2020.
- [7] Bharath Chintagunta, Namit Katariya, Xavier Amatriain, and Anitha Kannan. Medically aware gpt-3 as a data generator for medical dialogue summarization. In Ken Jung, Serena Yeung, Mark Sendak, Michael Sjoding, and Rajesh Ranganath, editors, *Proceedings of the 6th Machine Learning for Healthcare Conference*, volume 149 of *Proceedings of Machine Learning Research*, pages 354–372. PMLR, 06–07 Aug 2021.
- [8] Sumanth Dathathri, Andrea Madotto, Janice Lan, Jane Hung, Eric Frank, Piero Molino, Jason Yosinski, and Rosanne Liu. Plug and play language models: A simple approach to controlled text generation. *arXiv preprint arXiv:1912.02164*, 2019.
- [9] Ion Stagkos Efstathiadis, Guilherme Paulino-Passos, and Francesca Toni. Explainable patterns for distinction and prediction of moral judgement on reddit. *arXiv preprint arXiv:2201.11155*, 2022.
- [10] Denis Emelin, Ronan Le Bras, Jena D Hwang, Maxwell Forbes, and Yejin Choi. Moral stories: Situated reasoning about norms, intents, actions, and their consequences. *arXiv preprint arXiv:2012.15738*, 2020.
- [11] Angela Fan, Yacine Jernite, Ethan Perez, David Grangier, Jason Weston, and Michael Auli. ELI5: long form question answering. *CoRR*, abs/1907.09190, 2019.
- [12] Angela Fan, Mike Lewis, and Yann Dauphin. Hierarchical neural story generation, 2018.
- [13] Anna Guimaraes, Oana Balalau, Erisa Terolli, and Gerhard Weikum. Analyzing the traits and anomalies of political discussions on reddit. *Proceedings of the International AAAI Conference on Web and Social Media*, 13(01):205–213, Jul. 2019.
- [14] Donghoon Ham, Jeong-Gwan Lee, Youngsoo Jang, and Kee-Eung Kim. End-to-end neural pipeline for goal-oriented dialogue systems using GPT-2. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 583–592, Online, July 2020. Association for Computational Linguistics.
- [15] Ethan Haworth, Ted Grover, Justin Langston, Ankush Patel, Joseph West, and Alex C Williams. Classifying reasonability in retellings of personal events shared on social media: A preliminary case

- study with/r/amitheasshole. In *Proceedings of the International AAAI Conference on Web and Social Media*, volume 15, pages 1075–1079, 2021.
- [16] Steven C. H. Hoi, Doyen Sahoo, Jing Lu, and Peilin Zhao. Online learning: A comprehensive survey. *CoRR*, abs/1802.02871, 2018.
 - [17] Liwei Jiang, Jena D. Hwang, Chandrasekhar Bhagavatula, Ronan Le Bras, Maxwell Forbes, Jon Borchardt, Jenny Liang, Oren Etzioni, Maarten Sap, and Yejin Choi. Delphi: Towards machine ethics and norms. *ArXiv*, abs/2110.07574, 2021.
 - [18] Jiwei Li, Michel Galley, Chris Brockett, Jianfeng Gao, and Bill Dolan. A diversity-promoting objective function for neural conversation models. north american chapter of the association for computational linguistics. 2016.
 - [19] Gary Marcus and Ernest Davis. *Rebooting AI: Building Artificial Intelligence We Can Trust*. Pantheon Books, USA, 2019.
 - [20] Elle O’Brien. Aita for making this? a public dataset of reddit posts about moral dilemmas. *DVC*, Feb 2020.
 - [21] Elle O’Brien. Aita for making this? a public dataset of reddit posts about moral dilemmas. 2020.
 - [22] Nicholas Proferes, Naiyan Jones, Sarah Gilbert, Casey Fiesler, and Michael Zimmer. Studying reddit: A systematic overview of disciplines, approaches, methods, and ethics. *Social Media and Society*, 7(2), 2021. Funding Information: Ongoing research in this space is funded in part by NSF award IIS1704369 as part of the PERVADE (Pervasive Data Ethics for Computational Research) project. Publisher Copyright: © The Author(s) 2021.
 - [23] Reddit. Am i the asshole.
 - [24] Zeerak Talat, Hagen Blix, Josef Valvoda, Maya Indira Ganesh, Ryan Cotterell, and Adina Williams. A word on machine ethics: A response to jiang et al. (2021), 2021.
 - [25] Jennifer Wortman Vaughan. Making better use of the crowd: How crowdsourcing can advance machine learning research. *Journal of Machine Learning Research*, 18(193):1–46, 2018.
 - [26] Steven Euijong Whang, Yuji Roh, Hwanjun Song, and Jae-Gil Lee. Data collection and quality challenges in deep learning: A data-centric ai perspective, 2021.
 - [27] Tianyi Zhang*, Varsha Kishore*, Felix Wu*, Kilian Q. Weinberger, and Yoav Artzi. Bertscore: Evaluating text generation with bert. In *International Conference on Learning Representations*, 2020.

10 Appendix

You chose Title 1.

Posted by u/wrathfulSalami6:

My son Mark is a medical student. Very proud of him. He worked hard for years and we supported him every step of the way.

He has a girlfriend, Sarah, who is to put it simply a brat. She's also a medical student, and when he brought her round here and we were chatting about what got her wanting to study medicine, she started going on about how more people would study medicine if Doctors were paid better. (They're paid pretty well in this country far as I'm aware) She was saying that if medicine was a business they'd be paid much more, and that having to pay for healthcare would give people an incentive to work hard.

As someone who grew up not very well off, I'm very proud of the free healthcare we have in this country, though I appreciate that it's not perfect, but this woman believes that we're better off without it simply so she can earn more money (in a profession that pays fairly well anyway). She was saying that people like ourselves and her parents shouldn't have the same healthcare as people who ''don't work as hard''.

Her views aren't the only reason I take issue with her. She's constantly calling him while he's trying to study. He's doing well but I'm worried his grades will slip because he keeps running to her. She's studying the same course as him and while he's said she's very capable, she doesn't seem to take her studies as seriously as he does since she's always dragging him out with her friends.

I just don't think she's a good influence on him at all. I appreciate that they're adults but I'm allowed to have an opinion, but he wasn't happy when I told him.<END>

=====

What a moral conundrum, you think that the wrathfulSalami6 is:

YTA

NTA

Figure 2: Example game interface of generated stories.

You chose NTA.
You feel uncertain about this decision! You decide to look at
some comments arguing against it to help you guide your decision.

u/awedLlama0:
YTA. You should not control your child

u/alertDotterel7:
Definitely YTA, can't believe this is happening in 21th century

u/gleefulBasmati9:
YTA, Why do you think you have right to choose your son's gilfriend?

=====

Ok, so that's what THEY think. But what I think is

YTA
NTA

Figure 3: Example game interface of generated comments.